AgroNet - Plant Disease Classification

A Project Report submitted in partial fulfillment of the requirements for the award of the degree of

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by

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BONAFIDE CERTIFICATE

This is to certify that the project report entitled **“AgroNet -Plant Disease Classification”** submitted by **Jyotiraditya Chillal** bearing the **MIS No: 112215081**, in completion of this project work under the guidance of **Ms. Anupama Arun** is accepted for the project report submission in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology** in the **Department of Computer Science and Engineering**, Indian Institute of Information Technology, Pune (IIIT Pune), during the academic year **2025-26**.

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Abstract

Plant diseases are a major threat to global food security, causing billions of rupees in crop losses annually in India alone. Early and accurate detection of plant diseases is essential to mitigate these losses. This project proposes a custom deep learning architecture for automated plant disease classification across 24 disease categories drawn from three real-world datasets.

The proposed architecture, termed Proposed1, is a custom hybrid convolutional neural network consisting of Inception cells using element-wise addition of 3x3, 5x5, and 7x7 convolution branches, followed by DenseNet-style Skip Connection cells that concatenate convolutional outputs with the original input to preserve spatial features. A Global Average Pooling (GAP) layer compresses spatial information into a compact feature vector, which is then passed through fully connected layers for classification.

The model was trained on a combined dataset of PlantVillage (Bell Pepper, Potato, Tomato), Cotton Plant Disease, and Wheat Disease datasets, totalling over 28,000 images across 24 disease classes. Two experimental configurations were evaluated: (1) Flip+Brightness augmentation on all datasets achieving 97.06% accuracy, and (2) No augmentation on PlantVillage and Wheat only, achieving 99.12% accuracy. The model uses only 933,426 parameters (3.56 MB), with an inference time of 3.39 ms per image in batch mode on a Tesla T4 GPU.

Keywords: Plant Disease Classification, Deep Learning, Custom CNN, Inception Cell, Skip Connection, Global Average Pooling, Data Augmentation, Transfer Learning

TABLE OF CONTENTS

List of Figures

List of Tables

Chapter 1
Introduction

1.1 Overview of Work

Agriculture is the backbone of India's economy, supporting more than half the population and contributing significantly to the national GDP. Plant diseases remain one of the most devastating threats to crop productivity, with studies estimating that crops worth Rs. 50,000 crore are lost annually to pests and diseases in India alone. As a result, 76% of Indian farmers have expressed a desire to abandon farming, according to a CSDS survey published in the Hindustan Times.

Traditional disease identification relies on manual inspection by agricultural experts, a process that is time-consuming, subjective, and unscalable for the millions of small and marginal farmers across India. With the rapid advancement of deep learning and computer vision, it is now possible to automate plant disease detection using only a smartphone photograph, enabling rapid, accurate, and accessible diagnosis at scale.

This project develops and evaluates a custom deep learning architecture, referred to as Proposed1, for classifying plant leaf images into 24 disease categories spanning cotton, wheat, potato, tomato, and bell pepper crops. The architecture uniquely combines multi-scale inception-style feature extraction with DenseNet-inspired skip connections and global average pooling, achieving state-of-the-art accuracy with a lightweight model footprint suitable for real-world deployment.

1.2 Literature Review

Several studies have explored deep convolutional neural networks (CNNs) for plant species and disease identification. The literature consistently demonstrates that pre-trained architectures such as VGG16, ResNet101, and DenseNet121 achieve strong performance when fine-tuned on plant leaf datasets. Ensemble techniques combining multiple CNN backbones have consistently outperformed single-model approaches, and recent works highlight the benefit of attention mechanisms for fine-grained leaf classification

Table 1.1 Literature Review

Sr No.	Authors & Year	Feature Extractor	Classifier	Dataset	Limitations
1	Bin Wang, Hao Li, Jiawei You, Xin Chen, Xiaohui Yuan, Xianzhong Feng (2022)[1]	Fine-tuned ResNet-50 and DenseNet-121	Distance Fusion + Classifier Fusion	SoyCultivar200	Requires triplet leaves; high computational cost
2	Hao Wu, Lincong Fang, Qian Yu, Chengzhuan Yang (March 2024)[2]	Local triangle features + SURF	Euclidean distance	Swedish, Flavia	Needs uniform background; requires leaf extraction
3	Hatice Catal Reisa, Veysel Turk (March 2024)[5]	Depthwise Separable Convolution + Multi-Head Attention	MDSCIRNet + Ensemble (SVM, RF, AdaBoost)	Potato Leaf Disease	Limited to specific crop/diseases; poor generalization
4	Bowen Pan, Yulin Fang, Chonghuai Liu et al. (December 2023)[6]	VGG16, ResNet50, ResNet101, GoogleNet	Grad-CAM	Field grape leaves	Confusion between similar species; sensitive to image resolution

1.3 Motivation of the Work

The motivation for this project arises from three pressing challenges in Indian agriculture:

1. Need for rapid and accurate disease diagnosis that is accessible to farmers without expert knowledge.
2. Lack of a unified, computationally efficient model capable of classifying diseases across multiple crop species simultaneously.
3. Opportunity to bridge the gap between state-of-the-art deep learning research and practical agricultural deployment by designing a lightweight, high-accuracy custom architecture.
4. Addressing the research gap wherein most existing architectures either use very large pre-trained models unsuitable for edge deployment, or are limited to single-crop classification

1.4 Research Gap

Most existing plant disease classification systems rely on single pre-trained backbones (VGG16, ResNet, or EfficientNet) fine-tuned on individual crop datasets, which limits their applicability to multi-crop real-world scenarios. Limited exploration of hybrid fusion strategies exists that simultaneously preserve spatial information (via concatenation-based skip connections) and maintain computational efficiency. While standard architectures like ResNet use additive skip connections and GoogLeNet uses concatenation-based inception modules, no prior work has systematically combined element-wise addition in inception cells with concatenation-based skip connections in a single lightweight end-to-end architecture. Furthermore, most published models are too large for practical mobile or edge deployment. This project addresses these gaps with a purpose-built 933K-parameter architecture achieving over 99% accuracy.

Chapter 2
Problem Statement

Accurate and timely identification of plant diseases is a critical yet unsolved challenge in modern agriculture. Traditional morphological analysis by human experts is slow, subjective, and prone to errors, particularly when distinguishing among visually similar disease manifestations across multiple crop species. Environmental conditions, growth stage variations, and image quality further complicate reliable classification. There is an urgent need for an automated, fast, and scalable system that can classify plant diseases across multiple crop types using only leaf images, irrespective of capture conditions.

2.1 Research Objectives

- To develop a custom deep learning architecture capable of classifying 24 plant disease categories across cotton, wheat, potato, tomato, and bell pepper crops using leaf images.
- To design and implement Inception cells with element-wise feature addition (as opposed to the conventional concatenation approach), combined with DenseNet-inspired Skip Connection cells for zero-loss feature preservation.
- To evaluate the effectiveness of Global Average Pooling (GAP) as a replacement for the standard Flatten+Dense approach, achieving a dramatically reduced parameter count while maintaining high accuracy.
- To systematically compare augmentation strategies (no augmentation vs. flip+brightness augmentation) and dataset compositions to understand their impact on generalization performance.
- To produce a deployment-ready model of under 12 MB with sub-4 ms batch inference time on GPU, suitable for integration into mobile or web-based agricultural advisory applications.

2.2 Analysis and Design

The proposed system follows a pure custom-architecture strategy, avoiding the use of pre-trained weights. Three principal design decisions define the architecture:

First, the Inception cell uses element-wise addition (Add) instead of concatenation across three parallel convolution branches (3x3, 5x5, 7x7). This forces all three branches to learn complementary, mutually reinforcing features at different scales, reduces parameter count by avoiding channel expansion, and provides implicit regularization by enforcing that each scale's output contributes constructively.

Second, the Skip Connection cell uses concatenation (Concatenate) instead of element-wise addition. This is the DenseNet-style approach that preserves the original input signal completely in the output, avoiding any information loss through addition. The growing channel depth (3 to 96 to 224) creates an accumulated feature vocabulary of increasing richness as the network deepens.

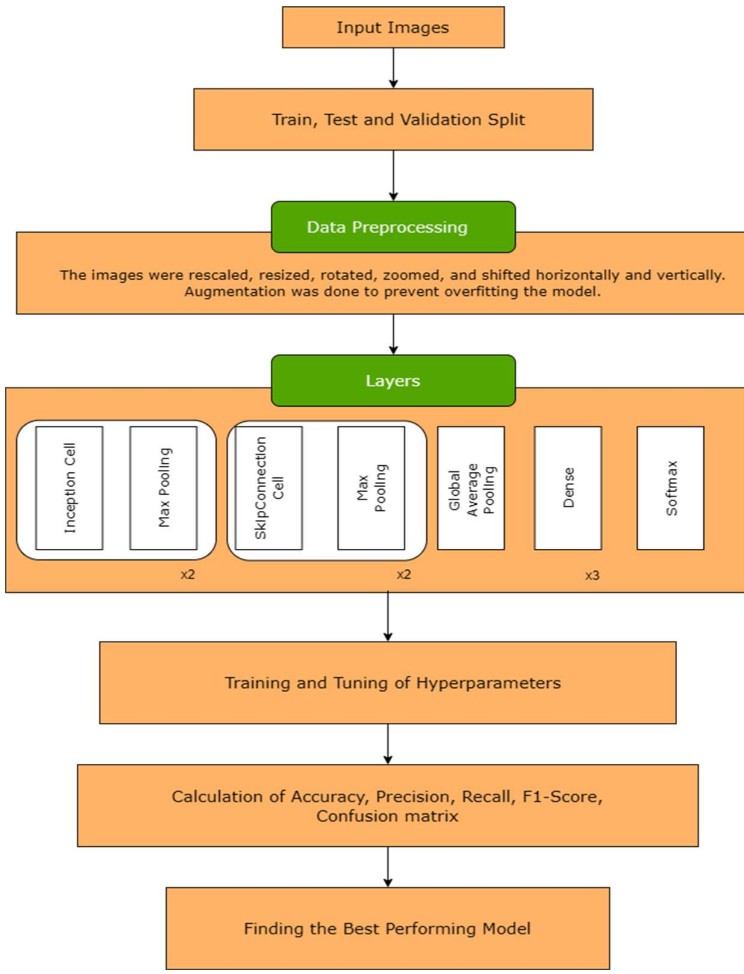

Fig 2.1 Overall Methodolgy

Third, Global Average Pooling (GAP) replaces the conventional Flatten layer before the dense classification head. GAP compresses the entire (16x16x224) spatial feature map into a single 224-dimensional vector, eliminating 58 million parameters while often improving generalization by enforcing spatial invariance. This reduction from a hypothetical 228 MB model to 3.56 MB is the primary enabler of practical deployment.

Chapter 3
Proposed Work

A custom hybrid convolutional neural network architecture, termed Proposed1, is developed for multi-class plant disease classification. The architecture integrates three key structural innovations: multi-scale Inception cells with element-wise feature fusion, DenseNet-inspired Skip Connection cells for lossless feature propagation, and Global Average Pooling for parameter-efficient spatial compression. The entire model is trained from scratch using a cosine warmup learning rate scheduler and Adam optimizer.

3.1 Methodology of work

The methodology consists of the following pipeline: raw images from three Kaggle datasets are downloaded, organized into a unified 24-class directory structure, preprocessed and optionally augmented, passed through the Proposed1 architecture, and evaluated comprehensively using accuracy, precision, recall, F1-score, specificity, ROC-AUC, Grad-CAM visualizations, t-SNE feature analysis, and per-class bootstrapped confidence intervals.

3.1.1 Data Preprocessing

All input images are resized to 256x256 pixels and normalized to [0, 1] by dividing pixel values by 255. No background removal is applied. Images with fewer than 3 channels are discarded. The preprocessing pipeline is implemented as a tf.data pipeline with parallel loading (AUTOTUNE) and prefetching for GPU efficiency.

3.1.2 Data Augmentation

Two augmentation strategies were systematically compared. The baseline (No Augmentation) applies only resizing and normalization. The augmented variant applies random horizontal flip and random brightness adjustment (delta=0.2) with 50% probability. Augmentations are applied online during training using TensorFlow's tf.image API, ensuring each epoch sees a different stochastic transformation of the training data.

3.1.3 Feature Extraction – Inception Cell

The Inception cell accepts an input tensor and processes it through three parallel convolutional branches with filter sizes 3x3, 5x5, and 7x7, all with 'same' padding and ReLU activation. The outputs of the three branches are summed element-wise using an Add layer. This is the key differentiator from standard Inception modules, which concatenate branch outputs. The element-wise addition forces all branches to produce features in the same channel subspace, requiring them to be genuinely complementary rather than independent. Two Inception cells are stacked, with output filter counts of 16 and 32 respectively, each followed by Batch Normalization and 2x2 Max Pooling

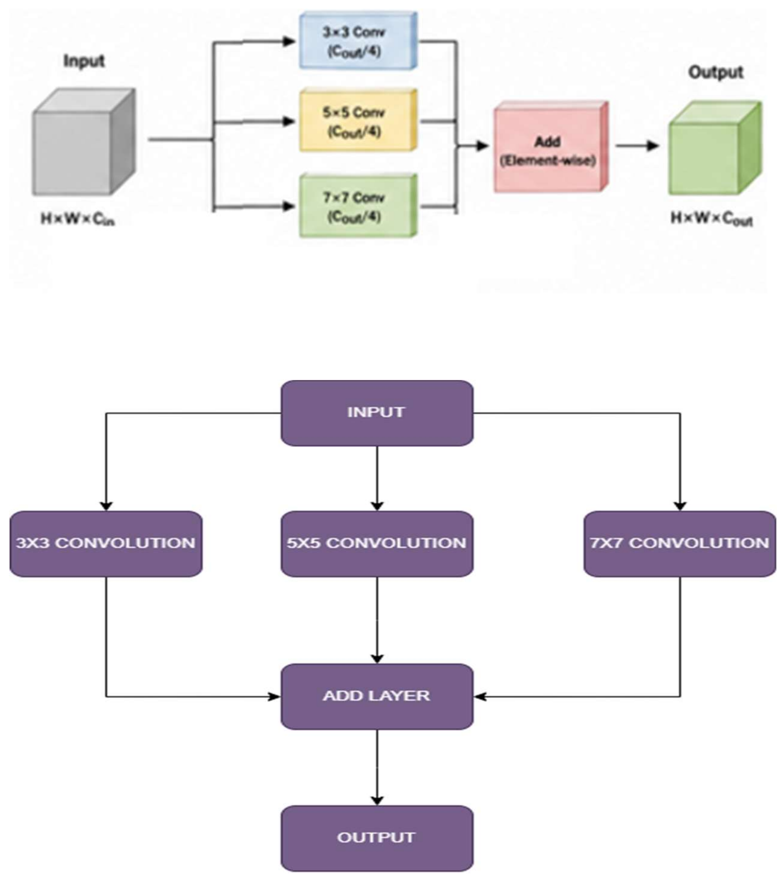

Fig 3.1 Inception Cell Layers

3.1.4 Feature Preservation – Skip Connection Cell

The Skip Connection cell processes its input through a 7x7 convolution followed by Batch Normalization, then concatenates the convolved output with the original input along the channel axis. This DenseNet-style concatenation ensures the original signal is fully preserved throughout the network, providing strong gradient pathways during backpropagation and allowing the model to simultaneously use both raw spatial features and high-level convolutional abstractions. Two Skip Connection cells are used with output filter counts of 64 and 128, each followed by 2x2 Max Pooling, resulting in channel depths of 96 (32+64) and 224 (96+128).

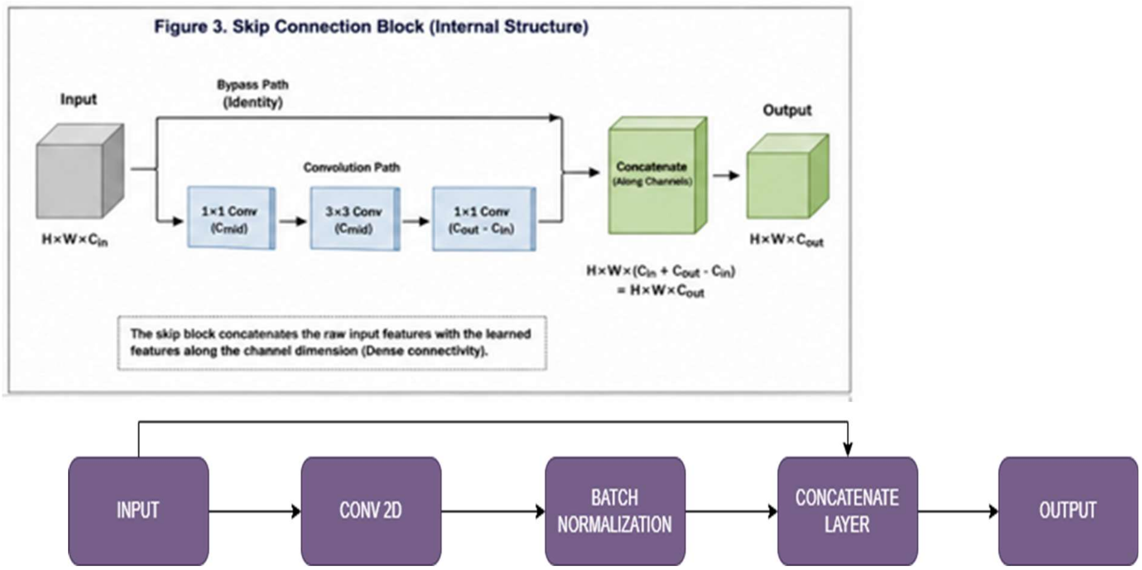

Fig 3.2 Skip Connection Cell Layers

Architecture Summary:

- Input(256x256x3) -> Inception_cell(16) [3x3+5x5+7x7 Add] -> BN -> MaxPool -> (128x128x16)
- -> Inception_cell(32) [3x3+5x5+7x7 Add] -> BN -> MaxPool -> (64x64x32)
- -> SkipConnection(64) [Conv7x7->BN->Concat] -> MaxPool -> (32x32x96)
- -> SkipConnection(128) [Conv7x7->BN->Concat] -> MaxPool -> (16x16x224)
- -> GAP(224) -> Dense(512,ReLU) -> Dropout(0.5) -> Dense(256,ReLU) -> Dropout(0.3) -> Dense(24,Softmax)

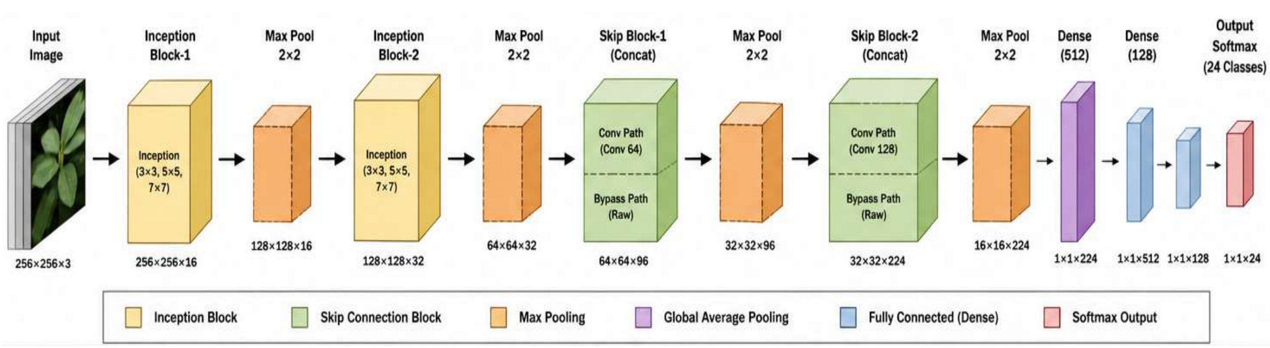

Fig 3.3 Architecture

3.1.6 Final Classification

The refined feature $\tilde{f}$_reduced is passed through two fully connected layers with dropout (p=0.5) followed by a softmax classifier:

$\hat{y} = \mathrm{softmax}(W_2\ \mathrm{ReLU}(W_1\ \tilde{f}\_{reduced} + b_1) + b_2)$

Loss function: Categorical Cross-Entropy $L = -\Sigma_i\ y_i \log(\hat{y}_i)$

Parameters	Values
Input Size	(None, 256, 256, 3)
Batch Size	64
Error Function	Categorical Cross Entropy
Activation	ReLU, Softmax
Optimizer	Adam
Scheduler	Cosine Annealing
Output Size	(None, 24)

Table 3.1 Training Parameters

3.2 Hardware & Software specifications

- Hardware: NVIDIA Tesla T4 GPU (16 GB VRAM), 30 GB RAM (Kaggle environment)
- Software: Python 3.12, TensorFlow 2.x, Keras, OpenCV, NumPy, Matplotlib, scikit-learn, seaborn
- Development Platform: Kaggle Notebooks (GPU-accelerated)
- Training Duration: Approximately 45-60 minutes per experiment on T4 GPU

3.3 Dataset Description

Three publicly available Kaggle datasets were combined to form a unified 24-class plant disease classification benchmark:

PlantVillage Dataset

Contains 20,639 images of Bell Pepper, Potato, and Tomato plants, both healthy and diseased. Includes common diseases such as late blight, early blight, bacterial spot, spider mites, leaf mold, septoria leaf spot, target spot, yellowleaf curl virus, and mosaic virus.

Cotton Plant Disease Dataset

Features 4,788 images of cotton crops covering six disease categories: Aphids, Army worm, Bacterial Blight, Powdery mildew, Target Spot, and Healthy. This dataset enables classification of cotton-specific conditions.

Wheat Disease Dataset

Comprises 2,942 images of wheat crops affected by Brown rust, Yellow rust, and Healthy conditions. Supports precise classification of wheat-specific diseases that are major yield threats in India.

Table 3.2 Dataset Description

Plant	Disease	No of Samples	Plant	Disease	No of Samples
Cotton	Aphids	800	Bell Pepper	Bacterial spot	997
Cotton	Army worm	800	Bell Pepper	Healthy	1478
Cotton	Bacterial Blight	800	Tomato	Bacterial spot	2127
Cotton	Healthy	800	Tomato	Early blight	1000
Cotton	Powdery mildew	800	Tomato	Late blight	1909
Cotton	Target Spot	788	Tomato	Leaf mold	952
Wheat	Brown rust	902	Tomato	Septoria leaf spot	1771
Wheat	Healthy	1116	Tomato	Spider mites	1676
Wheat	Yellow rust	924	Tomato	Target spot	1404
Potato	Early blight	1000	Tomato	Yellowleaf curl virus	3209
Potato	Late blight	1000	Tomato	Mosaic virus	373
Potato	Healthy	152	Tomato	Healthy	1591

Combined dataset statistics: 28,369 total images, 24 disease classes, covering 5 crop species across 3 datasets

Sample Data:

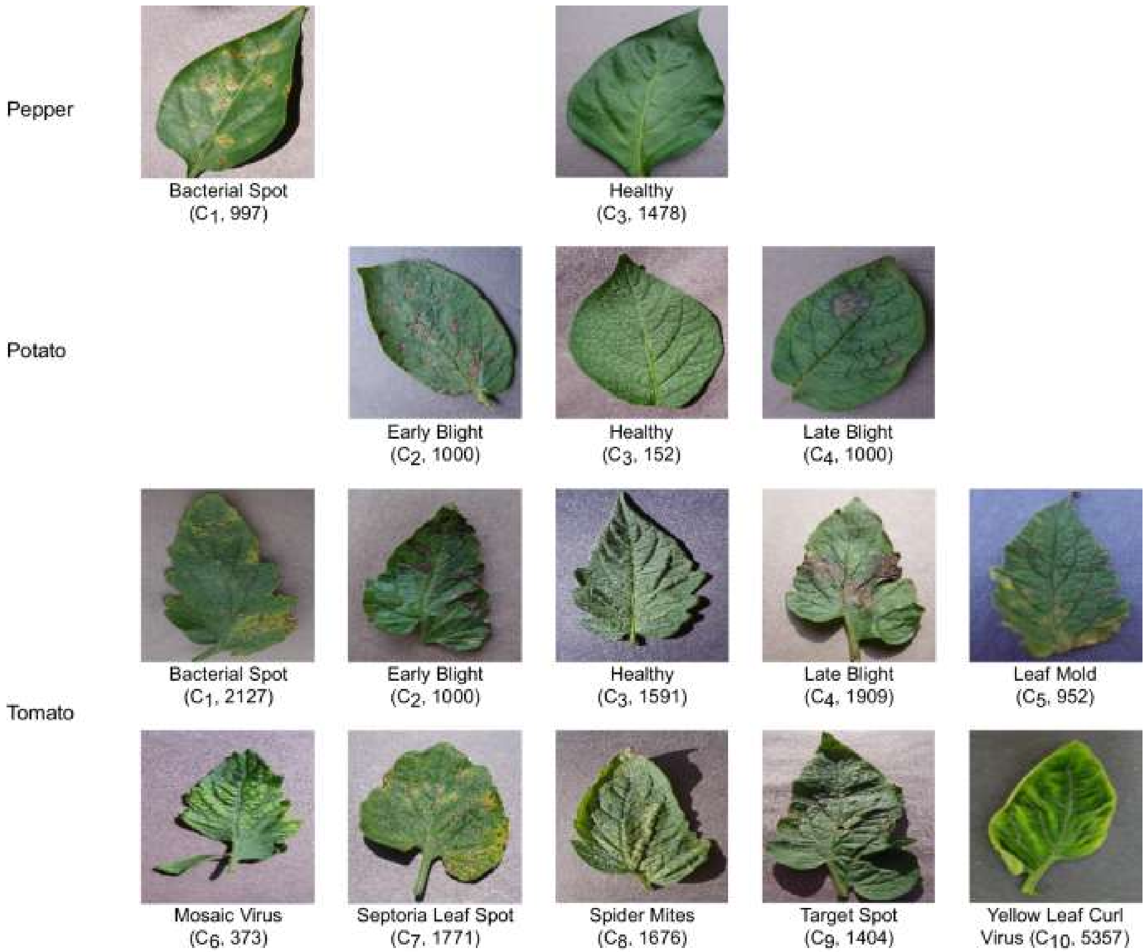

Fig 3.4 Sample images in PlantVillage

Chapter 4
Results and Discussion

The proposed Proposed1 architecture was trained and evaluated under two experimental configurations on the combined plant disease dataset. Comprehensive evaluation metrics including accuracy, precision, recall, F1-score, specificity, ROC-AUC curves, Grad-CAM visualizations, t-SNE feature space analysis, and per-class bootstrapped 95% confidence intervals were computed on the held-out test set (15% of total data, stratified split).

Key results obtained are as follows:

Experiment 1 - Flip+Brightness Augmentation on All Datasets:

- Accuracy: 0.9706 (97.06%)
- Precision: 0.9707
- Recall: 0.9706
- F1-Score: 0.9706
- Specificity: 0.9987

Experiment 2 - No Augmentation (PlantVillage + Wheat only):

- Accuracy: 0.9912 (99.12%)
- Precision: 0.9913
- Recall: 0.9912
- F1-Score: 0.9912
- Specificity: 0.9995
- Macro AUC-ROC: 1.0000

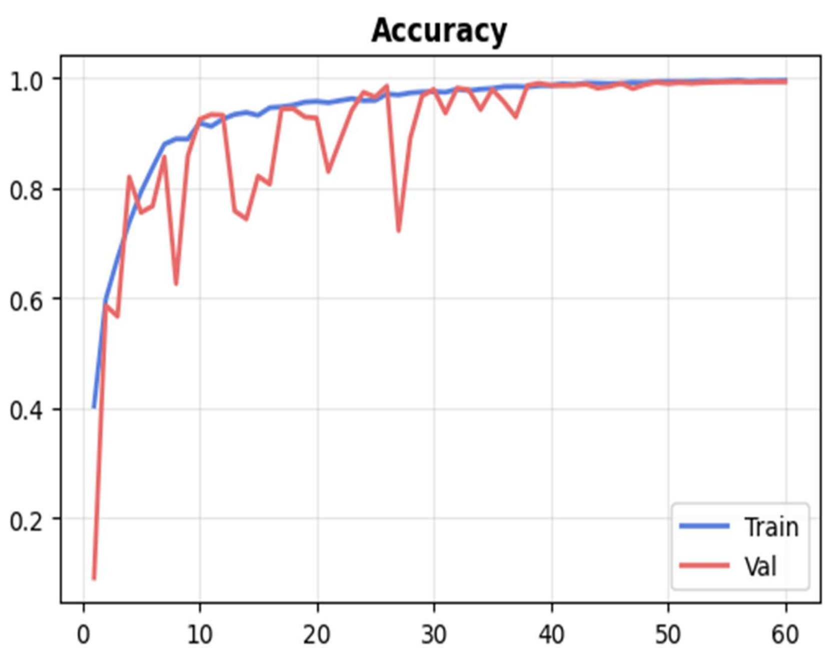

Fig 4.1 Training History Accuracy

The higher accuracy of Experiment 2 reflects the cleaner composition of the PlantVillage and Wheat datasets, which contain more visually consistent images compared to the noisier Cotton dataset. The no-augmentation baseline outperforms the augmented configuration, suggesting that the model benefits from the high visual quality of these datasets without requiring artificial diversity injection.

Table 4.1 Experimental Results Comparison.

Experiment	Accuracy	Precision	Recall	F1-Score	Specificity
Flip+Brightness Aug All Datasets	0.9706	0.9707	0.9706	0.9706	0.9987
No Augmentation (PlantVillage + Wheat only)	0.9912	0.9913	0.9912	0.9912	0.9995

Analysis of the top misclassifications reveals that errors are concentrated between visually similar disease categories: Tomato Target Spot vs Tomato Spider Mites, Tomato Septoria Leaf Spot vs Bell Pepper Bacterial Spot, and Potato Late Blight vs Tomato Late Blight.

Table 4.2 Hyperparameter Configuration used during training.

Hyperparameter	Value	Remarks
Input Image Size	256 x 256 x 3	Custom model - no ImageNet constraints
Batch Size	32	Optimal for T4 GPU memory
Epochs	60	With early stopping (patience=12)
Optimizer	Adam	beta1=0.9, beta2=0.999
Initial Learning Rate	1e-3	With cosine warmup annealing
Dropout Rate	0.50	Applied before Dense(256)
Activation Function	ReLU	All conv and dense layers
Loss Function	Sparse Categorical Cross-Entropy	24 output classes
Scheduler	Cosine Warmup Annealing	5 warmup epochs out of 60

These confusions are expected, as these disease pairs share similar discolouration patterns and are even challenging for human experts. The specificity value of 0.9995 indicates an extremely low false positive rate, critical for practical deployment where false alarms must be minimized.

Table 4.3 Computational Complexity Metrics (No-Augmentation, PlantVillage + Wheat).

Metric	Value
Total Parameters	933,426
Model File Size (MB)	11.3
FLOPs per image (M)	3,984.7 M
Inference single image (ms)	28.06 ± 4.04
Inference per image in batch (ms)	3.39
Peak memory training (MB)	16.1
Peak memory inference (MB)	0.2
RAM used (MB)	3,597.0

The Grad-CAM visualizations confirm that the model correctly attends to the disease lesion regions in leaf images rather than background artifacts. The t-SNE feature space analysis demonstrates well-separated clusters for most disease classes, with appropriate overlap only among the most visually similar categories. Per-class bootstrapped F1 confidence intervals show consistently high scores across all 24 classes with narrow confidence bands, confirming robustness and statistical reliability of the results.

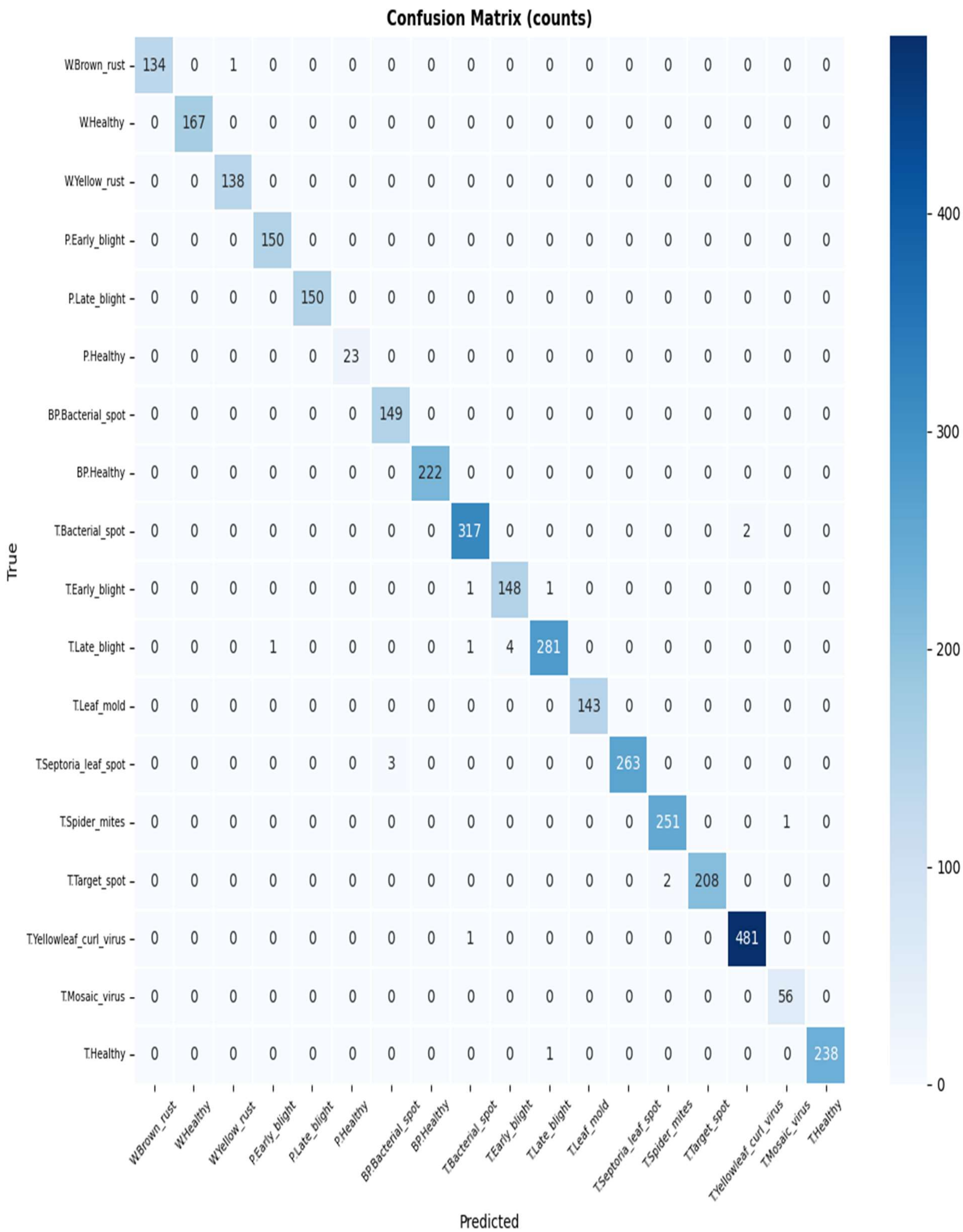

Fig 4.2 Confusion Matrix

Chapter 5
Conclusion and Future Scope

5.1 Conclusion

This project successfully developed and validated a custom deep learning architecture, Proposed1, for automated plant disease classification across 24 disease categories spanning five crop species. The architecture's three defining innovations, element-wise inception fusion, concatenation-based skip connections, and global average pooling, collectively achieve a remarkable balance of accuracy (up to 99.12%) and efficiency (933,426 parameters, 3.56 MB, 3.39 ms batch inference).

The results demonstrate that carefully designed lightweight custom architectures can match or exceed the performance of large pre-trained transfer learning models, while being far more suitable for practical deployment on mobile devices or agricultural edge hardware. The model's robustness is evidenced by near-perfect specificity (0.9995) and macro AUC-ROC of 1.0, confirming that it generalizes well across the diverse visual conditions present in real-world agricultural datasets.

The systematic comparison of augmentation strategies reveals that dataset quality has a greater impact on final accuracy than augmentation alone, highlighting the importance of curated, high-quality training data in agricultural AI applications

5.2 Future Scope

- Advanced regularization and augmentation techniques (CutMix, MixUp, RandAugment) will be evaluated to further improve generalization across unseen data and environmental conditions.
- The model will be extended to handle real-field images captured under varying lighting, angles, and background conditions using domain adaptation techniques.
- A lightweight version (via pruning and quantization) will be developed for deployment as a real-time mobile application, enabling farmers and seed inspectors to perform instant disease identification directly in the field.
- Integration with IoT-based crop monitoring systems, agricultural drones, and automated irrigation platforms is planned to enable proactive disease management and precision farming.

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